Q1. What is AI in Data Visualization? Explain how it differs from traditional data analysis.

Answer :- AI in Data Visualization refers to the integration of machine learning (ML) and natural language processing (NLP) to automate the discovery and communication of data patterns. Unlike traditional data analysis, which is descriptive and relies on a human analyst to manually build charts and find trends, AI-driven visualization is proactive. It can automatically surface hidden correlations, predict future outcomes, and allow users to "ask" questions of their data in plain English.

Q2. Explain how AI helps in automatic insight generation in BI tools.

Answer :- Modern BI tools use AI to scan datasets for anomalies, trends, and clusters without manual intervention. This is achieved through:

- **Pattern Recognition: Identifying significant deviations or spikes in data.**
- **Natural Language Generation (NLG): Translating complex data points into human-readable summaries.**
- **Automation: Reducing the time spent on "data slicing" by instantly highlighting the most relevant drivers of a specific metric.**

Q3. What are the key AI features available in Microsoft Power BI? Explain any two with examples.

Answer :- Power BI offers several AI-specific visuals and features. Two key examples include:

- **Q&A Visual: This allows users to ask questions using natural language.**
    - ***Example:* Typing "What were the total sales in New York last month?" and having Power BI automatically generate a bar chart.**
- **Decomposition Tree: This visual lets users drill into different dimensions to see what contributes to a high or low value.**
    - ***Example:* If "Profit" is low, you can use the tree to see which specific product category or region is responsible for the dip.**

Q4. Explain the concept of Key Influencers and how AI identifies factors affecting a metric.

Answer :- The Key Influencers visual uses machine learning (specifically regression and classification) to analyze your data and rank the factors that have the greatest impact on a selected metric.

- **How it works: If you want to know what drives "Customer Churn," the AI compares customers who left versus those who stayed, identifying which variables (like contract type or monthly charges) most likely influenced the outcome.**

Q5. What is the role of Smart Narratives in AI-powered dashboards?

Answer :- Smart Narratives provide a text-based summary of the visuals on a dashboard. Their role is to:

- **Improve Accessibility: Helping users quickly understand the "so what?" of a chart.**
- **Contextualize Data: Automatically updating the text as users apply filters. For instance, if you filter by "Year 2024," the text summary will instantly update to describe the trends for that specific year.**

Q6. Explain how Tableau Pulse provides proactive insights in Tableau.

Answer :- Tableau Pulse is a reimagined data experience that uses generative AI to push insights to users where they work (like Slack or Email).

- **Personalization: It learns which metrics a user cares about most.**
- **Proactivity: Instead of waiting for a user to open a dashboard, it alerts them when a metric (like "Inventory Levels") deviates from the norm and explains *why* it happened using "Next-Best-Action" suggestions.**

Q7. What is meant by AI-driven decision-making? How does AI support business decisions?

Answer :- AI-driven decision-making involves using data-backed predictions rather than just intuition.

- **Support Mechanism: AI supports decisions by providing Predictive Analytics (what will happen?) and Prescriptive Analytics (what should we do about it?). This reduces human bias and allows for faster responses to market changes.**

Q8. Explain the importance of interpreting AI-generated insights instead of directly trusting them.

Answer :- While AI is powerful, it lacks contextual business knowledge.

- **Correlation vs. Causality: AI might find a correlation between two variables that is actually a coincidence.**
- **Bias: If the underlying data is biased, the AI's "insight" will be biased too. Human interpretation acts as a "sanity check" to ensure the insight is ethically sound and practically applicable.**

Q9. Discuss the limitations of AI in data visualization tools.

Answer :-

- **Data Quality: AI is only as good as the data it consumes ("Garbage in, Garbage out").**
- **Lack of "Why": AI can tell you *what* is happening, but it often struggles to explain the external human factors (like a global pandemic or a sudden policy change) that caused the shift.**
- **Complexity: Advanced AI features can sometimes produce "Black Box" results that are difficult for non-technical users to explain to stakeholders.**

Q10. Compare how AI is used differently in Power BI and Tableau.

Feature	Microsoft Power BI	Tableau
Integration	Deeply integrated with the Office 365 ecosystem and Azure ML.	Deeply integrated with Salesforce (Einstein Discovery).
User Focus	Focused on "Self-Service" AI with easy-to-use visuals like Q&A.	Focused on "Augmented Analytics" and deep exploratory data science.

Language	**Uses "Q&A" for natural language queries.**	**Uses "Ask Data" and "Tableau Pulse" for AI-driven summaries.**